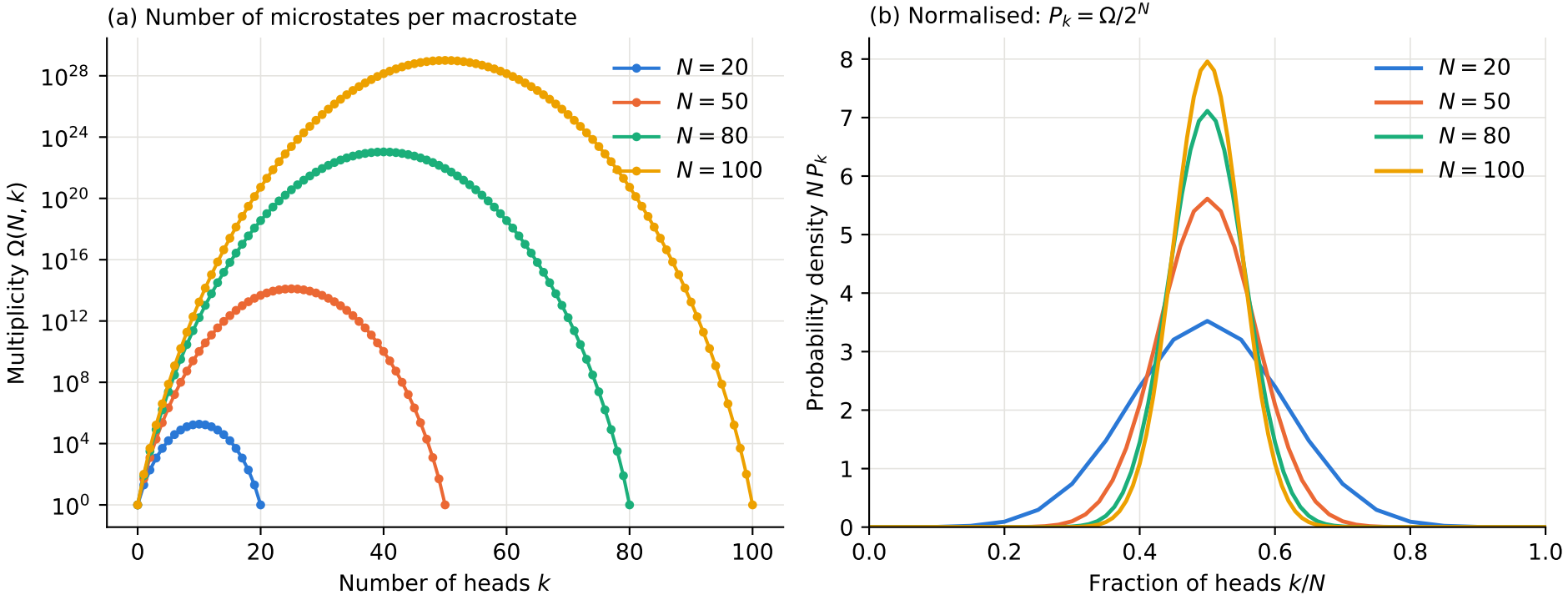
(a) Number of microstates per macrostate
Multiplicity $\Omega(N, k)$
10^{28}
10^{24}
10^{20}
10^{16}
10^{12}
10^{8}
10^{4}
10^{0}
0
20
40
60
80
100
Number of heads k
$N = 20$
$N = 50$
$N = 80$
$N = 100$
(b) Normalised: $P_k = \Omega/2^N$
Probability density NP_k
0
1
2
3
4
5
6
7
8
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of heads k/N
$N = 20$
$N = 50$
$N = 80$
$N = 100$